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28 November 1966
SUPERSEDING
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MILITARY SPECIFICATION

TEST STAND, AIRCRAFT HYDRAULIC SYSTEM TTU-228/E, TRAILER-MOUNTED, GASOLINE-ENGINE-DRIVEN, TRIPLE SYSTEM

1. SCOPE

1.1 This specification covers a triple-system (each system providing 20 gpm at 3,000 psi and 10 gpm at 5,000 psi), trailer-mounted, aircraft hydraulic system test stand, designated TTU-228/E.

2. APPLICABLE DOCUMENTS

*2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-G-76	Gages, Pressure and Vacuum, Dial Indicating, (for Air, Steam, Oil, Water, Ammonia and Chloro-fluoro Hydrocarbon Gases)
UU-T-81	Tags, Shipping and Stock
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproofing, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-1137	Varnish, Electrical-Insulating (for Electromotive Equipment)
MIL-F-3541	Fittings, Lubrication
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design, Installation, and Data Requirements for
MIL-R-5520	Reservoirs, Aircraft, Hydraulic, Non-Separated Type
MIL-H-5593	Hose, Aircraft, Low-Pressure, Flexible
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservative and Testing
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-E-10062	Engines: Preparation for Shipment and Storage of
MIL-E-16298	Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of
MIL-C-20696	Cloth, Coated, Nylon, Waterproof

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MIL-E-26248	Engines, Gasoline, Ground Use, Packette, General Specification for
MIL-F-27656	Filter, Fluid, Pressure, Absolute 5 Micron, Hydraulic

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS24217	Cable Assembly-Power, Electrical, 28 Volt DC Power
MS24560	Engine-Gasoline, Type GLU-4/U 175 HP Packette (formerly PE150)
MS28720	Filter, Aircraft Hydraulic Line Type, 3,000 PSI
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic and Pneumatic (3,000 PSI), Flared Tube
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MS51335	Pintle Assembly, Towing, 18,000 lbs. Capacity, Manual Release

Air Force-Navy Aeronautical

AN832	Union-Flared Tube 3/8 Bulkhead and Universal
AN894	Bushing-Screw Thread Expander
AN910 AN919	Coupling - Pipe Thread Reducer - External Thread Flared Tube
AN924	Nut - Flared Tube, Bulkhead and Universal Fitting
AN929	Cap Assembly, Pressure Seal Flared Tube Fitting
AN2551	Cable Assembly, Power, Electrical, 28 Volt DC
AN3150	Battery, Storage, Shielded, 24-Volt, 36-Ampere-Hour
AN6235	Filter Element-Hydraulic Replaceable Micronic Line Type
AN6236	Filter Element-Hydraulic Replaceable Micronic Reservoir Type
AN6249	Valve, 3,000 PSI Hydraulic Check
AN6270	Hose Assembly-Detachable Swivel Fitting, Low Pressure

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AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND10051	Bosses-Conduit Connection, Standard Dimensions for
AND10266	Flange-Type XVI Accessory Mounting
AND20006	Drive-Type XVI Engine Accessory

DRAWINGS

Air Force

48B7796	Ring Assembly-Tie Down, 10,000 Lb
49B6564	Ring-Eyebolt, 25,000 Lb Cargo Tiedown
49B7292	Receptacle-Battery, Quick Disconnect
49B7293	Plug-Battery, Quick Disconnect

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The test stand shall consist of the following major components:

Item or System Name	See Requirement
Hydraulic system	3.9
Gasoline engine system	3.10
Air compressor	3.11
Fan or blower	3.12
Electric power harness	3.13
Trailer and housing	3.14.

3.3 General specifications. The requirements of the following specifications, as specified, apply as requirements of this specification with the exceptions and additions specified herein. When the general specifications and this specification conflict, this specification shall govern.

a. MIL-H-5440	Requirements for materials, fluids, reservoir, venting, tubing, and cleaning of parts and systems
b. MIL-M-8090	General requirements and those specified for type II, ground C mobility.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials

3.5.1 Protective treatment. When materials are used in the construction of the test stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective

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coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

* 3.6 Design. The test stand shall be designed with three independent systems providing a total output of not less than 60 gpm at 3,000 psi and 30 gpm at 5,000 psi. Each system shall deliver 20 gpm at 3,000 psi and 10 gpm at 5,000 psi when operated as a three-system stand. Each system shall deliver 30 gpm at 3,000 psi and 15 gpm at 5,000 psi per system when operated as a two-system stand.

3.6.1 Arrangement of components. The components of the test stand shall be arranged as shown on figures 1 and 2. The components shall be arranged for maximum ease of operation and maintenance consistent with the versatility and efficiency required to conform to the requirements specified herein.

3.6.2 Reliability. The equipment shall have a minimum MTBF (mean-time-between failures) of 152 hours with a minimum reliability of 0.967 for a 5-hour mission at a confidence factor of 0.90 (see 6.4.1 and 6.4.2).

3.6.3 Maintainability. The test stand shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity

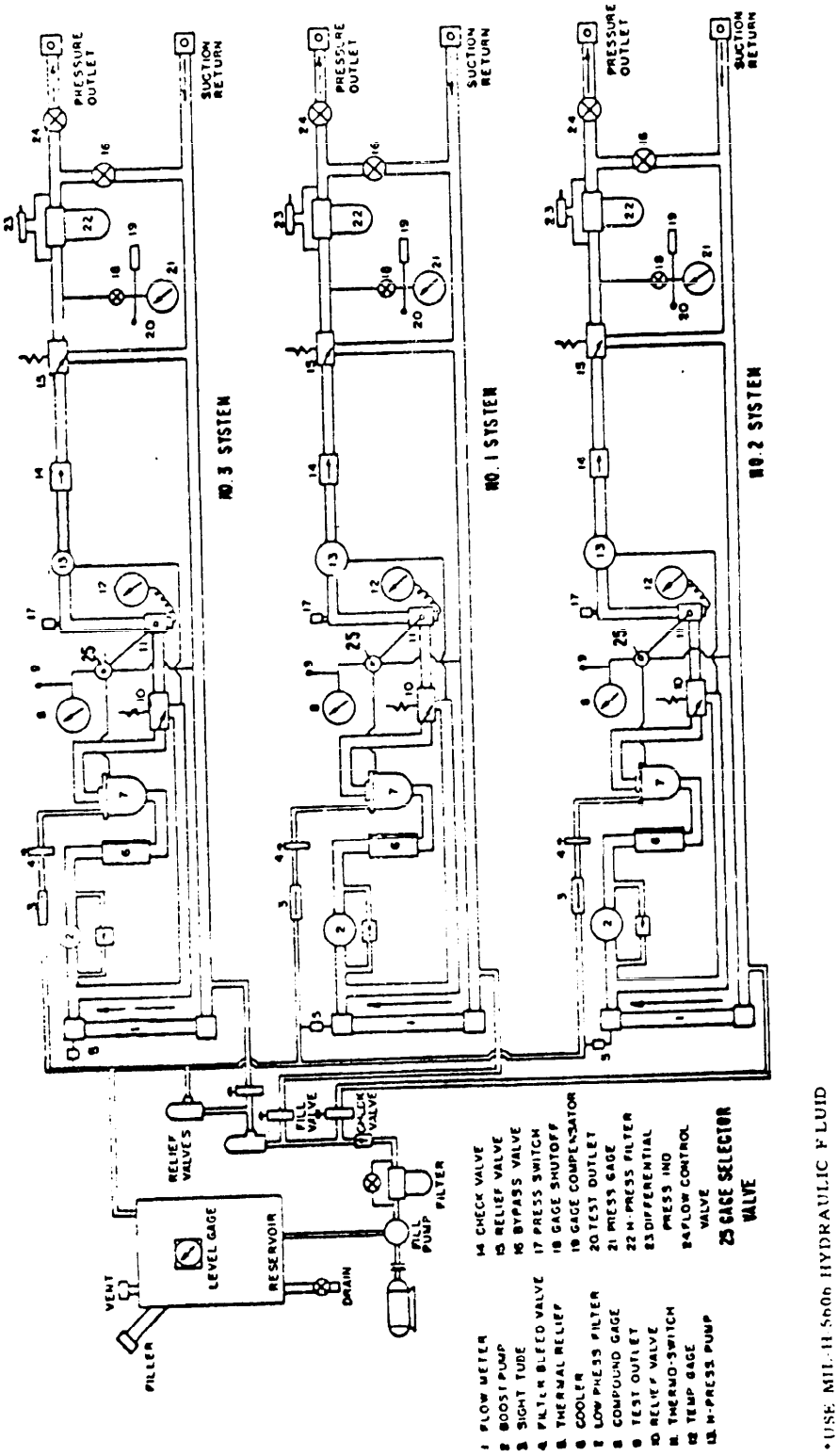
e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolthead sizes to the practicable minimum.

3.6.3.1 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided where practical. This mounting shall be so designed that the item can be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.7 Construction. The test stand shall withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.7.1 Safety. The test stand shall be constructed to provide for maximum

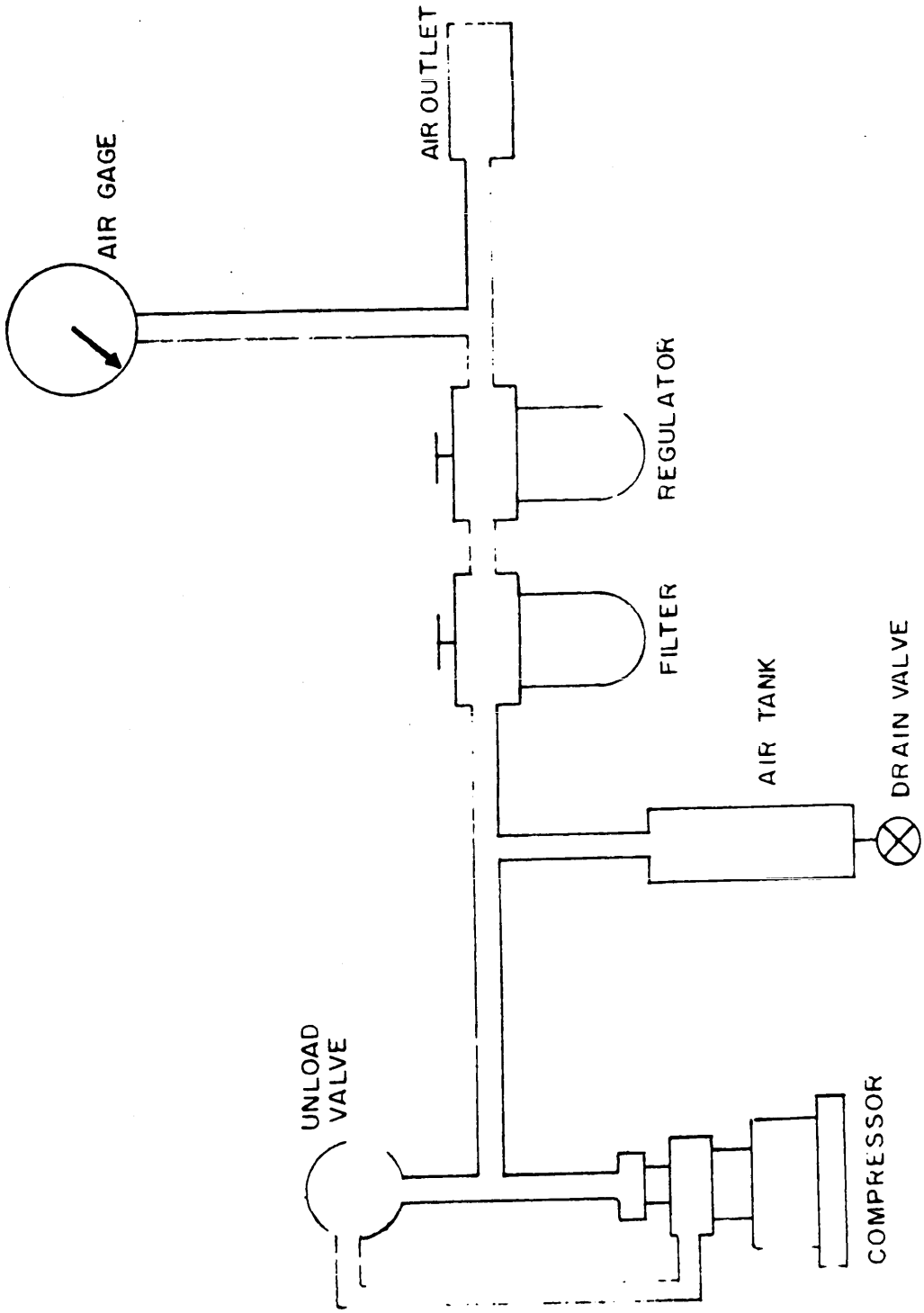
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*USE MIL-H 5606 HYDRAULIC FLUID

Figure 1. Schematic of Hydraulic System

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★Figure 2. Pneumatic System

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safety to operating or maintenance personnel.

3.7.2 Operating and service controls. Operating and service controls shall be readily accessible to and operable by an operator wearing heavy gloves.

3.7.3 Fittings. All fittings and threaded bosses in the hydraulic circuit and components shall be as specified in AND10050 or MS33656. All 5,000-psi pressure fittings shall be steel. Bulk-head-type, high-pressure fittings shall use a locknut with a backup gasket.

3.8 Performance. The test stand shall operate under and satisfactorily resist the following:

a. Temperatures ranging from -40° to +125°F

b. Pressure altitude from sea level to 6,000 feet

c. Fungus growth as encountered in tropical climates

d. Sand and dust particles as encountered in desert areas

e. A relative humidity of 95 to 100 percent at an operating temperature of 125°F and a storage temperature of 160°F

f. Exposure to salt-fog

g. Rainfall as encountered in any locale.

3.8.1 Storage temperatures. The test stand shall withstand storage temperatures ranging from -65° to +160°F.

3.8.2 Operating positions. The test stand shall function satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from true horizontal.

3.9 Hydraulic system

3.9.1 High-pressure pumps. Three axial-piston-type, high-pressure pumps shall be provided. They shall have aircraft pump impulse characteristics by using nine pistons and high operational speeds (1,500 to 4,000 rpm inclusive). The pumps shall be of the variable-stroke, variable-pressure, pressure-compensated type with adjustable maximum volume control. Each pump shall be of adequate rating to meet the requirements specified herein, but shall be not less than 30 gpm at 3,000 psi and 15 gpm at 5,000 psi at 3,750 rpm.

3.9.1.1 Life expectancy. The high-pressure pumps shall have a minimum life expectancy of 1,000 hours of continuous duty at full rated load.

3.9.1.2 Each pump shall require not more than 40 bhp from the gasoline engine specified in 3.10 when continuously pumping hydraulic fluid conforming to MIL-H-5606 at the rate of 20 gpm and raising the fluid pressure from 110 to 3,000 psia at 68°F.

3.9.1.3 Each pump shall operate satisfactorily throughout inlet-fluid temperatures ranging from -40° to +250°F and pressures from 110 to 265 psig.

3.9.1.4 Each pump shall have an adjustable pressure-compensator control capable of regulating the pressure at a constant value with the flow varying from 0 to 30 gpm. The discharge pressure of the pumps shall be 2,700 ±50 psi when delivering 30 gpm and when the zero flow pressure setting is 3,000 psi. The pump shall meet the performance characteristics shown on figures 3 and 4 at any speed above 50 percent of rated speed. The pump shall limit the maximum

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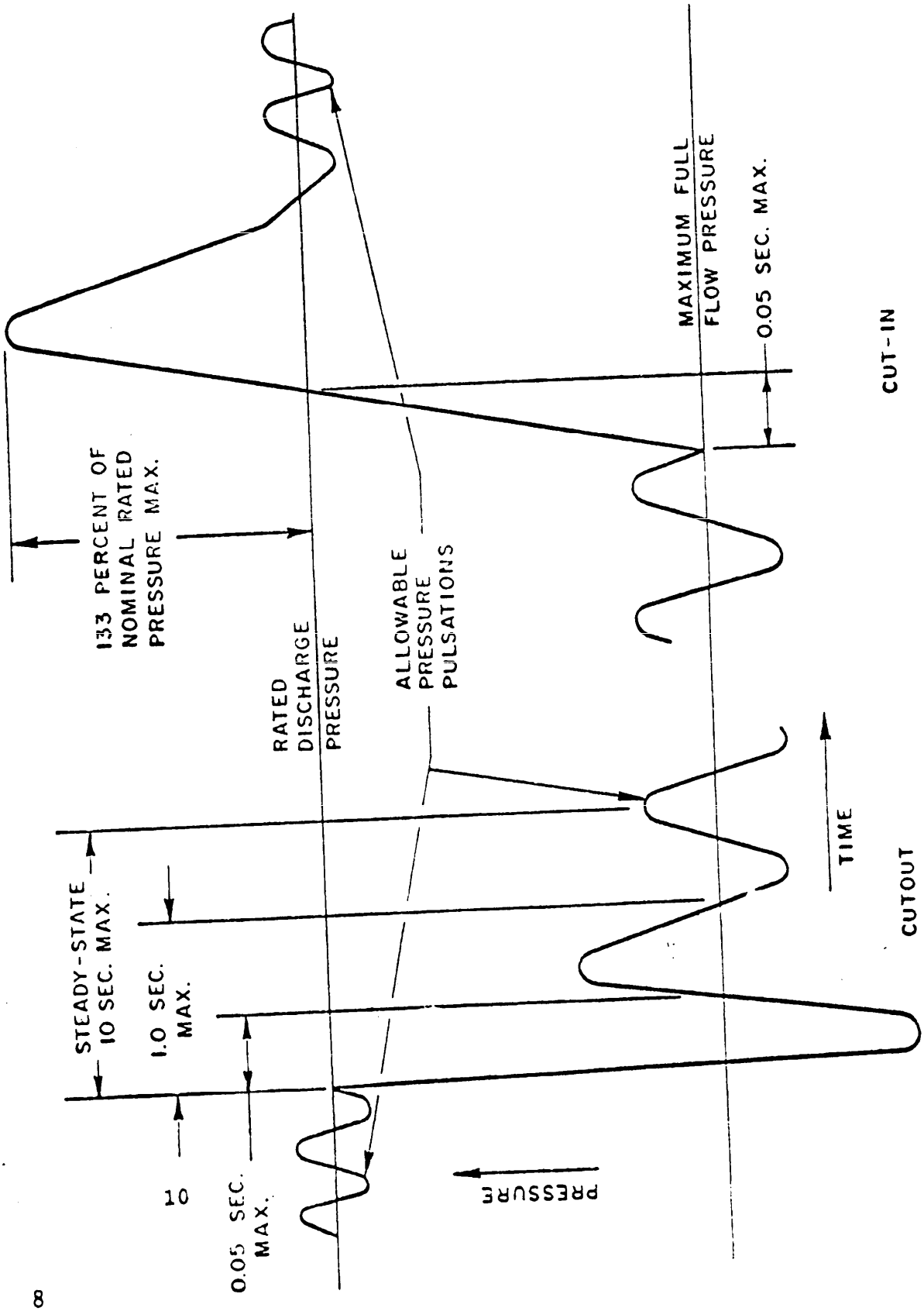


Figure 3. Typical Pressure Vs Timechart

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CAPACITY SHALL BE WITHIN THE LIMITS SHOWN WHILE OPERATING AT 3,750 RPM AND USING HYDRAULIC FLUID AT 140° F

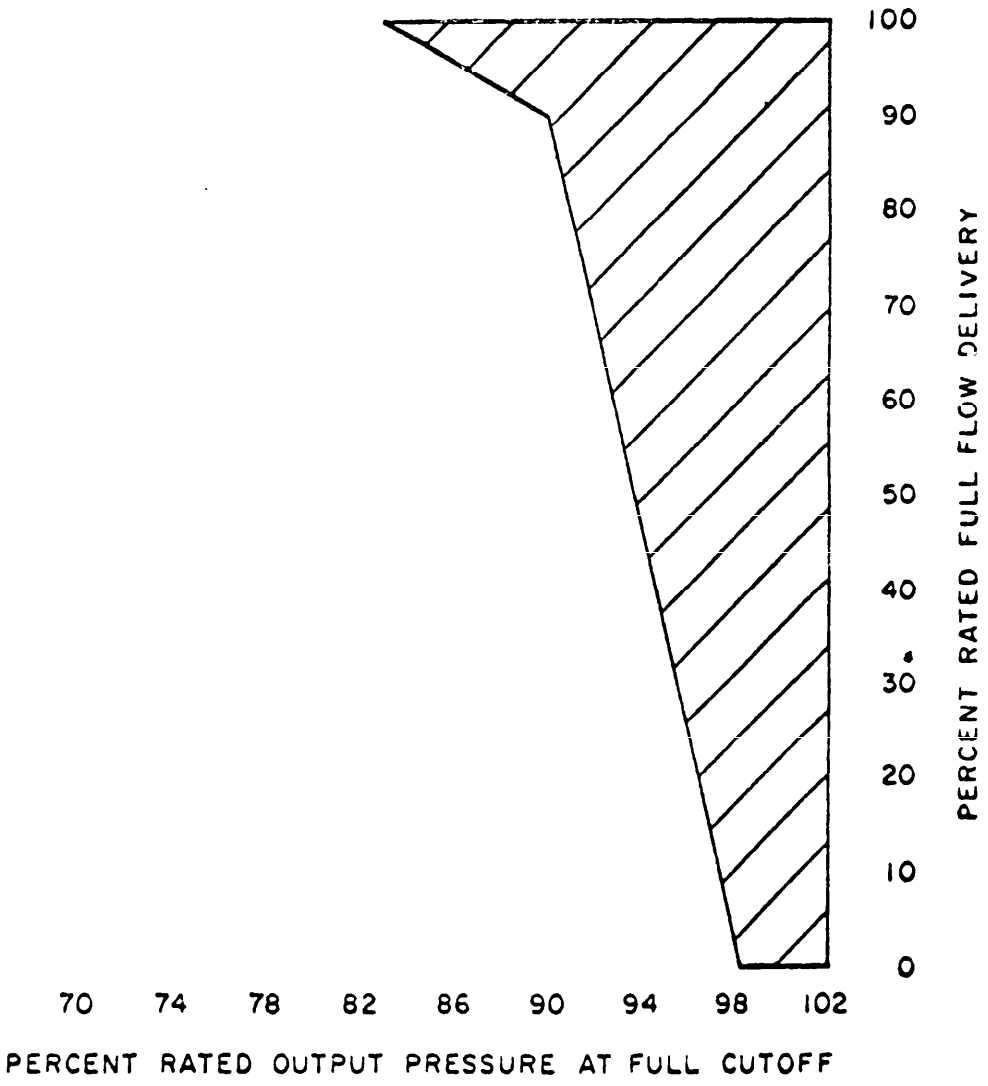


Figure 4. Flow Characteristics

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transient pressure during cutout to not more than 133 percent of nominal rated pressure. The adjustment range of the pressure compensator shall be from 500 to 5,000 psi. The variable volume control and pressure control shall be mounted on the control panel.

3.9.1.5 A fixed relief valve shall be incorporated to maintain the pump case pressure at a value recommended by the pump manufacturer.

3.9.16 The pumps shall be furnished with flanges and a spline drive conforming to AND10266, type XVI-C.

* 3.9.2 Low-pressure boost pumps. Three low-pressure hydraulic boost pumps shall be provided to meet the minimum life expectancy requirements specified in 3.9.1.1 Each boost pump shall supply at least 35 gpm continuously to each high-pressure pump with the inlet pressure at 20 inches Hg absolute and the outlet pressure not less than 10 psig nor more than 300 psig, as recommended by the high-pressure pump manufacturer, at all flow and pressure conditions specified herein.

3.9.2.1 Check valves. Check valves shall be provided and located as shown on figure 1 to allow fluid to bypass the boost pumps. The check valves shall be of the 1-1/2-inch, quick-acting, poppet type with a maximum of 2 psi pressure drop when passing 30 gpm.

3.9.3 High-pressure filters. Three high-pressure filters conforming to MIL-F-27656 shall be incorporated as shown on figure 1.

3.9.3.1 Differential pressure indicators. Differential pressure indicators shall be connected into the system as shown on figure 1 and shall indicate when the pressure drop across any filter is 100 ±10 psi. The indication shall be the closing of a switch which will illuminate red warning lights mounted in a conspicuous location on the instrument panel.

3.9.4 Low-pressure filters. Suction line filters shall be provided and located as shown on figure 1. Each filter shall utilize a minimum of two replaceable AN6236-3 or four replaceable AN6236-2 filter elements. The low-pressure filters shall be so located on a level above any other part of the line between the boost pump and the high-pressure pump that the air will collect within the filter and will not be discharged into the high-pressure pump inlet line.

3.9.4.1 Drain plugs. Drain plugs shall be provided. Sufficient clearance space shall be permitted around each item for draining and replacing the filter element.

3.9.4.2 Filter bleed valves. Push-button-type bleed valves shall be mounted on the control panel and shall serve to bleed entrapped air to the reservoir from the top of the low-pressure filter case or any other location at which air may be more apt to collect. The valves shall also serve to bleed or drain the reservoirs of the aircraft, when desired. Force necessary to operate this valve shall not exceed 10 pounds. When in the released position, the valve shall not allow leakage from the aircraft hydraulic system to exceed 2 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch.

3.9.4.3 Sight tubes. Sight tubes shall be furnished in the bleed lines as shown on figure 1. They shall be mounted on the control panel adjacent to their respective

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bleed valves. The sight tubes shall withstand a 200-psi working pressure and a 300-psi proof pressure. They shall be 3/8-inch size and 3-inch minimum visible length. The sight tubes shall be lighted from behind.

*3.9.4.4 Thermal relief valves. Thermal relief valves shall be provided to protect the test systems from overpressurization due to expansion of fluid when heated. The valves shall bypass fluid to the test stand reservoir when the fluid pressure exceeds 150 psig.

3.9.5 Relief valves. High- and low-pressure relief valves shall be located in the system as shown on figure 1. The relief valves shall be located as near as practical to the high-pressure pump inlet and outlet.

3.9.5.1 High-pressure relief valves. High-pressure relief valves shall be adjustable by hand from 500 to 5,000 psig. The high-pressure relief valves shall be mounted on the control panel and shall incorporate a locking device so that vibrations cannot change the settings.

3.9.5.2 Low-pressure relief valves. The low-pressure relief valves shall be adjusted to a fixed pressure of not less than 110 psi nor more than 265 psi. The low-pressure relief valves shall be sensitive to back-pressure. They shall close to zero flow within 0.025 second and shall have a pressure drop no greater than 25 psi when passing 35 gpm.

3.9.6 Pressure outlets. Three pressure outlets shall be provided as shown on figure 1. The outlets shall not be located where leakage of fluid from the outlet could fall on the trailer tires or within the housing of the test stand. The outlets and dust caps shall be equal to and interchangeable with Aeroquip parts No. 015628-S2-16 and 155-S7-16D respectively. Each outlet shall be color-coded and labeled PRESSURE OUTLET NO. _____ by an engraved or etched plastic or metal nameplate, permanently attached.

*3.9.6.1 Pressure connections. All valves, connections, and tubing used from the high-pressure pump discharge to the end of the 30-foot pressure hoses (each consisting of three lengths of hose listed in table I) shall be of a size that will not allow pressure drop to exceed 160 psi with 30 gpm flow. All connections and fittings on the high-pressure side shall be not less than 1 inch tubing size. Tubing shall be used on all high-pressure lines within the test stand, except where impractical because of vibration.

3.9.7 Return connections. Three suction connections shall be provided as shown on figure 1. The connections shall conform to Aeroquip part No. 145-S2-24D, or be equal to and interchangeable with the Aeroquip part. A dust cap conforming to Aeroquip part No. 145-S7-24D, or equal to and interchangeable with the Aeroquip part, shall be attached to each connection. The connections shall be located for convenience. Location shall not allow leakage from the connections to fall on the trailer tires nor within the test stand enclosure. Each return connection shall be color coded.

3.9.7.1 Low-pressure components and connections. All connections used in the suction side of the hydraulic system shall be of such size as to allow not more than a 10-psi drop at 30 gpm with fluid temperatures of 60°F from the aircraft outlets through 10 feet of hose to the suction side of the boost pumps. Suction lines and components shall be not less than 1-1/2-inch tubing size.

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TABLE I

HOSE ASSEMBLIES

Hose Assembly Part No.	Quantity	Hose Ends (One or Both)	Equipped With
MS28759-16-1200	3 ea	Both	*Coupling half, Aeroquip part No. 015828-S5-16 *Dust plug, Aeroquip part No. 155-S9-16D
MS28741-24-1200	3 ea	Both	*Coupling half, Aeroquip part No. 145-S5-24D *Dust plug, Aeroquip part No. 145-S9-24D
MS28759-16-1200	6 ea	One	*Coupling half, Aeroquip part No. 015630-S2-16 *Dust plug, Aeroquip part No. 155-S7-16D
		Other	*Coupling half, Aeroquip part No. 015828-S5-16 *Dust plug, Aeroquip part No. 155-S9-16D
MS28741-24-1200	6 ea	One	*Coupling half, Aeroquip part No. 145-S5-24D *Dust plug, Aeroquip part No. 145-S9-24D
		Other	*Coupling half, Aeroquip part No. 014519-S2-24D *Dust cap, Aeroquip part No. 145-S7-24D

*Or a part equal to and interchangeable with the Aeroquip part.

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3.9.7.2 External and internal leakage. Total external leakage shall not exceed 2 drops per hour at any flow or pressure specified herein. Internal leakage shall not exceed 5 cubic inches per hour from the closed loop system of either system.

*3.9.8 Oil cooler. An air-to-oil oil cooler(s) shall be provided in the systems as shown on figure 1. The oil cooler(s) shall maintain the oil temperature of each system below 140°F when 30 gpm at 750 psig is passed through the coolers with an ambient temperature of 100°F. They shall allow full flow through the bypass without damage resulting therefrom. The oil cooler(s) shall operate under a pressure of 200 psi without leakage or damage. The oil cooler(s) shall be provided with a means to make them inoperable in low temperature. Provisions for accomplishing the requirements specified herein shall be accessible, but shall not be on the control panel.

3.9.9 Flow control and bypass valves. Three flow control valves and three bypass valves of 1 inch or larger shall control or stop the flow into or from each pressure outlet. The valves shall be designed for 30 gpm flow and 5,000 psi pressure. They shall accurately meter the flow throughout the volume range of 0 to 30 gpm and shall be so pressure-balanced that a torque of 40 pound-inches is the maximum required turning force at 3,000 psi. The valves shall go from the closed position to the open position in two full turns of the handle. The valves shall control the flow within ±1/4 gpm when flow rates are below 5 gpm, and ±1 gpm with flow rates from 5 to 30 gpm.

3.9.10 Fill system. A system incorporating an electric-motor-driven pump, filters, pushbutton-type fill valves, and relief valves as shown on figure 1 shall be provided to allow 5-minute periods of continuously filling or adding oil to the systems as desired.

3.9.10.1 Filling pump. The filling pump shall be a low-pressure hydraulic pump having a capacity of at least 2 gpm at 150 psi that is driven by a direct-current motor powered by the engine battery. The pump shall be operable for at least 15 minutes without discharging the engine battery below the charge required to start the engine. During the 15-minute operation, the engine shall not be running. The pump motor shall be rated for continuous duty. A manual switch shall be provided on the instrument panel for running the pump motor.

3.9.10.2 Filling valves. Three pushbutton-type filling valves shall be mounted on the control panel. The valves shall be so interconnected that pushing the button to fill the system will also start the pump motor and releasing the button will stop the motor. Force necessary to operate the valve shall not exceed 10 pounds. When in the released position, the valves shall not allow leakage from the aircraft hydraulic system to exceed 10 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch.

3.9.10.3 Fill system relief valves. Two variable relief valves as shown on figure 1 shall be provided, but not mounted on the control panel. The valves shall be adjustable throughout a pressure range from 0 to 140 psi and relieving the total volume discharged by the fill pump. The valve shall have a locking device so that vibrations cannot change the pressure setting. Each relief valve shall be set for 110 psi.

3.9.10.4 Fill system check valve. A check valve shall be provided in the fill

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system as shown on figure 1 to prevent reverse flow to the fill system pump.

*3.9.10.5 Fill system filter. A filter shall be provided in the fill system as shown on figure 1. The filter shall be an AN6235-2A filter element in a filter assembly similar to an MS28720-6 filter suitable for 145 psi operating pressure, except with no bypass relief valve.

3.9.10.6 Fill system connections. Lines, tubing, hoses, and connections used in the fill system shall be 3/8-inch tubing size, or larger.

*3.9.11 Pressure check valves. Pressure check valves that are dimensionally similar to AN6249-16 valves, but suitable for operation at 5,000 psi, shall be located as near the high-pressure pump outlet as practical to prevent reverse flow to the high-pressure pumps. (See figure 1.)

3.9.12 Flow meters. Three flowmeters of the transparent type having a scale length of approximately 250 mm, a capacity flow of not less than 30 gpm, and minimum readable capacity down to 3 gpm shall be provided. The flowmeters shall be near or on the control panel and appropriately lighted. The flowmeters shall be accurate within ±2 percent of full-scale reading. The flowmeters shall be plain, tapered tube types with a rod guide for the float and shall be capable of operative pressures to 250 psi. The flowmeters shall be easily removable for cleaning without affecting calibration.

*3.9.13 High-pressure gages. Flush-mounted, corrosion-resistant, temperature-compensated, high-pressure gages conforming to GG-G-76 shall be provided. (Where conflict exists between this specification and GG-G-76, this specification shall govern.) The gages shall have a 6-inch diameter dial and a steel-alloy bourdon tube. The pointer shall be adjustable. The gages shall be suitable for operation at -40° to +200°F.

3.9.13.1 Mounting. All gages shall be of the recessed type provided with a mounting flange at the front of the gage. The mounting flange and dial shall extend not more than 5/8 inch beyond the external surface of the panel. All gages shall be mounted by screws and tapped panel holes to permit removal from the front of the panel by one man.

3.9.13.2 Range and accuracy. The gages shall provide a range of 0 to 6,000 psi and shall incorporate a means to restore the accuracy of the gages under all service conditions. The gages shall be accurate to within 1/2 percent, or less, of full scale.

3.9.13.3 Movements. Good-quality, stainless-steel movements shall be provided on all gages.

3.9.13.4 Dampener. Gages subjected to continuous surges and extreme pressure fluctuations shall be protected by a hydraulic dampener.

3.9.13.5 Safety features

3.9.13.5.1 Blowout device. All gages shall incorporate a safety blowout device. The device shall automatically relieve the gage case pressure should it reach 1/2 psi due to gage failure.

3.9.13.5.2 Wall. The dial and pointer shall be separated from the gage movement by a wall which will prevent rupture through the front of the gage in the event of gage failure.

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3.9.13.6 Connection. All fluid gages shall be connected to the panel with flexible AN- or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

3.9.13.7 Throttling. All pressure gages shall be equipped with needle valves, located as close as practicable to the gage.

3.9.13.8 Gage lines. The high-pressure gage lines shall be provided with needle shutoff valves, accumulator-type pulsation dampeners, and gage test fittings.

3.9.13.8.1 Shutoff valves. The needle shutoff valves shall be of the panel-mounted type with corrosion-resistant steel stems. The manufacturer's cold-working pressure rating shall be stamped on the valve body with an arrow indicating the flow direction. The valves shall be designed for working pressures up to 5,000 psi.

3.9.13.8.2 Gage test fittings. Gage test fittings shall be provided for the high- and low-pressure gages. The following fittings shall be mounted on the control panel adjacent to their respective gages:

a. An AN832-4D nipple

b. An AN924-4D nut

c. An AN929-4 cap.

3.9.14 Low-pressure gages. Flush-mounted compound gages having corrosion-resistant-steel bourdon tubes and a dial of 3-1/2-inch diameter, or larger, shall be provided. The gages shall be accurate to within 1 percent of full scale range and shall conform to GG-G-76, except where conflicts exist, the provisions of this specification shall govern. The gages shall provide a range of 30 inches Hg vacuum to 300 psi. The gage movements shall be constructed of stainless-steel and shall be designed for operation from -40° to +125°F. In all other respects, the gages and mounting shall conform to 3.9.13 through 3.9.13.8.2.

3.9.14.1 Gage selector valves. Three shutoff-type selector valves shall be mounted on the instrument panel and shall allow for pressure selection at the various points as shown on figure 1.

3.9.15 Temperature gages. Remote-reading temperature gages shall be provided. The range of the gages shall be approximately -20° to +200°F and shall indicate within ±5°F of the temperature of the fluid supply to the high-pressure pumps. The gages shall have a 3-inch diameter, or larger, dial and shall be flush-mounted. The dial faces of the temperature gages shall be marked with a red sector as a warning at 160°F and higher. The accuracy of the gages shall not be affected by continued exposure to the climatic extremes specified herein.

3.9.16 Duplex gage. A corrosion-resistant duplex gage with a 2-1/2-inch dial face shall be mounted on the stand panel to simultaneously indicate the pressure on both sides of the filter in the fill system. The duplex gage shall have a range from 0 to 200 psi, and shall be constructed of corrosion-resistant material throughout.

3.9.17 Thermoswitches. Adjustable type preset thermoswitches shall sound

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the warning horn when the oil inlet temperature to either high-pressure pump exceeds 165°F. The thermoswitches shall have an adjustable range from 150° to 200°F.

3.9.17.1 Horn. A horn shall be provided which shall sound a warning signal when the boost pressure falls below the recommended pressure, or set operation is beyond the recommended high temperature range. The horn shall be a 24-V, vibrator type and shall be audible at 25 feet from the test stand when the test stand is operating at full capacity.

3.9.18 Hose assemblies. The hose assemblies specified in table I shall be provided with the test stand and equipped as specified.

3.9.18.1 Dust plugs and caps. All dust plugs and caps shall be attached to the test stand or hose assemblies by metal chains to prevent loss.

3.9.19 Adapters. The following adapters shall be furnished in the stowage compartment:

3.9.19.1 Three each coupling halves, conforming to Aeroquip part No. 015630-S2-16 and No. 155-S5-12D, or halves equal to and interchangeable with the Aeroquip parts, shall be fastened into three double-threaded couplings similar to the couplings conforming to AN910 with AND10050-16 threads in one end and AND10051-12 threads in the other. The threaded couplings shall be not more than 3 inches long and shall be designed for working pressures of 5,000 psi and proof pressures of 7,500 psi.

3.9.19.2 Three each coupling halves, conforming to Aeroquip parts No. 014519-S2-24D and 145-S5-16D, or halves equal to and interchangeable with the Aeroquip parts, shall be fastened into three double-threaded couplings similar to the couplings conforming to AN910 with AND10050-24 threads in one end and AND-10050-16 threads in the other. The threaded couplings shall be not more than 3 inches in length and shall be designed for working pressures of 200 psi and proof pressures of 300 psi.

3.9.20 Accessories and loose fittings. The following accessories shall be provided and stored in the enclosed accessory box specified in 3.15:

Item	Part Number	Quantity
Reducer-External Thread Flared Tube	AN919-27D (-24 by -16)	3
Bushing Unit	AN894-16-8 (-16 by -8)	3
Coupling Half	*Aeroquip 150-S5-8D	3
Dust Plug	*Aeroquip 155-S9-8D	3

*Or a part equal to and interchangeable with the listed part.

3.10 Gasoline engine system

*3.10.1 Gasoline engine. An MS24560-1 engine shall be provided (see 3.24). The engine shall be mounted within the test stand housing.

3.10.1.1 Capacity. The horsepower input required by all items driven by the engine shall not exceed 75 percent of the full throttle rating of the engine at the

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selected operating speed and maximum temperature specified herein. While at 6,000 feet altitude, the test stand shall pump fluid at 3,000 psi and 20 gpm in the three systems simultaneously without overloading the engine.

3.10.2 Gasoline engine accessories. In addition to other items as may be necessary to meet the requirements specified herein, the following items shall be provided:

3.10.2.1 Fuel pump. A positive-displacement-type fuel pump shall be provided and shall have sufficient capacity to supply fuel to the engine at its maximum rated load when lifting the fuel against a suction head of 4 feet.

3.10.2.2 Governor. A governor shall be provided and shall automatically control the engine speed from a minimum practical speed at hydraulic pressures below 750 psi and the selected rated speed at hydraulic pressures above 750 psi. The manually-operated hand throttle shall control the engine at all speeds from the manufacturer's recommended idling speed up to the minimum operating speed extablished by the governor.

3.10.2.3 Starting and charging systems. An electrical starting system, manual starting system, and a 25-amp, battery-charging system with relay shall be provided and shall conform to MIL-E-26248.

3.10.2.4 Batteries. Two 24-V batteries conforming to AN3150 shall be provided for engine electric starting, operation of the fill motor, to provide external power, and for heater operation. The batteries shall be connected in parallel. Battery quick-disconnects conforming to Drawings 49B7292 and 49B7293 shall be installed for the batteries.

3.10.2.5 Fuse or circuit breakers. Fuse or circuit breakers shall be installed, as required, to protect the electrical system and wiring against faults and overload currents.

3.10.2.6 Fuel tank. A fuel tank of adequate capacity to provide fuel for 4 hours or more of normal load operation shall be mounted within the outline of the housing. The capacity of the fuel tank shall be not less than 42-1/2 gallons.

*3.10.2.6.1 Filler opening. A filler opening of not less than 2 inches inside diameter in accordance with MIL-R-5520 shall be located to facilitate filling from 5-gallon containers. The fuel tank cap shall be painted red and securely attached to the fuel tank by a metal chain.

3.10.2.6.2 Sump. A sump shall be provided in the fuel tank and shall be suitable for the collection and accumulation of not less than 1 pint of water.

3.10.2.6.3 Drain. A drain shall be located at the bottom of the sump to permit complete drainage of the fuel tank. Fittings shall carry the drainage from the fuel tank to the outside of the stand housing.

3.10.2.6.4 Fuel level gage. A fuel level gage shall be provided to indicate the quantity of available fuel in the tank. The gage shall be accurate to within ±2 gallons of actual quantity. The gage need not be on the control panel, but shall be easily visible to an operator standing at the control panel.

3.10.2.7 Driven components. The driven components shall be driven by a gear-type speed changer attached to the output end of the engine. Power from the

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engine to the speed changer and from the speed changer to the driven components shall be transmitted by means of flexible couplings or splined shafts. The drive pads for the high-speed pumps shall be in accordance with AND20006.

3.10.2.7.1 Clutches. A clutch shall be provided for each pump drive to connect and disconnect it from the driving train. The clutch controls shall be easily accessible, but need not be on the control panel. If the clutches are of the type that might be damaged by engaging them while the stand is running, they shall be protected from such damage. The clutches may be of the overriding type, if required, to prevent reverse rotation of the pumps in event of shutdown due to loss of boost pressure (see 3.9.2.1).

3.10.2.8 Heater. A heater shall be provided and shall preheat the engine and other cirtical components under sustained ambient temperatures from 0° to -40°F. After 30 minutes or less of preheating, successful engine starting shall be possible from a 24-V battery source of 68 ampere-hour capacity. The heater operation shall be thermostatically controlled after initial firing to maintain by cycling any heat necessary for either the engine or other critical components. If the heater requires electrical power, the average power consumption shall not exceed 2 ampere-hours at 24V for preheating. In addition to furnishing heat for the engine, the heater shall be provided with a special outlet for supplying warm air to the batteries used in conjunction with engine starting. Means shall be provided to protect the batteries from overheating. The heater shall not present a fire hazard. The fuel used for the heater shall be from the same supply as used for operating the engine.

3.10.3 Instruments and controls. The engine shall be provided with a manual throttle, manual choke, starter switch, tachometer, hourmeter, oil pressure gage, head temperature gage, oil temperature gage, ammeter, heater controls, and such other instruments or controls as may be required. All engine and heater instruments and controls shall be grouped together on one section of the instrument panel.

3.11 Air compressor. A commercial air compressor shall supply the air for pressurizing the aircraft reservoir. The compressor shall provide 3 cfm at 150 psi continuously when supplied with air at 77°F and 29.92 inches Hg absolute. An AN832-4D nipple, AN924-4D nut, and an AN929-4 cap shall be provided for the outlet connection. An outlet shutoff valve shall be provided to cut off air to the outlet connection. An air receiver, pilot-operated unloader valve, and 5-micron filter shall be provided in the pneumatic system as shown on figure 2. The air receiver shall be equipped with a moisture drain shutoff valve.

3.11.1 Pressure regulator. A commercial pressure regulator modified for satisfactory operation from -40° to +125°F ambient temperature shall be furnished. The pressure regulator shall control and limit the pressure to the outlet at any setting from 5 to 150 psi and shall maintain the set pressure at any flow condition including zero flow. A pressure gage shall be included to indicate pressure in the aircraft reservoir and shall be accurate to within 5 percent of full scale range.

3.11.2 Hose. One 30-foot length of AN6270-4 hose in accordance with MIL-H-5593 shall be provided.

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3.12 Fan or blower. Fan(s) or blower(s) with a drive mechanism from the gasoline engine or a hydraulic motor drive shall be furnished. The fan(s) or blower(s) and drive assembly shall withstand all conditions specified herein. The fan(s) or blower(s) shall be used with the air-to-oil cooler(s) and shall be located as shown on figure 1.

3.13 Electric power harness. A 30-foot-long electric power harness conforming to MS24217, with the plug wired in accordance with AN2551, shall be provided and installed on the test stand connected to the batteries of the engine starting system. A spring-return reel shall be provided for storing the electrical cable.

3.14 Trailer and housing. The components of the test stand shall be mounted and assembled on a trailer and covered with a metal weather-resistant housing. The complete test stand shall meet the requirements of MIL-M-8090 as specified for type II, group C mobility and shall be as follows:

3.14.1 Trailer

3.14.1.1 Frame. The trailer frame shall be of welded-steel construction. The frame shall be not less than 54 inches nor more than 66 inches wide and shall be not less than 84 inches nor more than 108 inches long. The ratio of width divided by length shall not exceed three-fourths.

3.14.1.2 Springs. Springs on the trailer shall provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 3 inches is added to the frame height for spring deflection.

3.14.1.3 Parking brakes. Brakes for both rear wheels shall be interconnected and shall operate from one handle.

3.14.1.4 Lifting and tiedown fittings. Lifting and tiedown fittings conforming to Drawing 48B7796 or 49B6564 shall be provided and shall be capable of withstanding the loadings specified in MIL-A-8421. The test stand shall be designed for forklifting.

3.14.1.5 Lubrication. All moving parts shall be provided with suitable means of lubrication and shall operate satisfactorily when lubricated with standard military lubricants as specified in MIL-M-8090.

3.14.1.5.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility. The test stand and all components shall be completely lubricated prior to delivery.

3.14.1.5.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached in a protected, conspicuous location.

3.14.1.6 Pintle hook. The test stand shall be equipped with a pintle hook conforming to MS51335 mounted on the center of the rear frame.

3.14.1.7 Control panel. A control panel shall be provided for mounting all

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meters, valves, indicators, gages, switches, and controls furnished with the stand. The control panel shall be constructed of No. 12 U.S. gage (0.1046 inch), or thicker, sheet steel and supported by a welded-steel, self-supporting framework. All instruments subjected to error from vibration shall be shock-mounted.

3.14.1.7.1 Location. The control panel frame shall be bolted or welded to the trailer bed and so positioned that the trailer wheels, tow bars, et cetera, will not interfere with the operator. The control panel shall be located at a sufficient height above the ground and inclined at a sufficient angle (10° minimum) to facilitate convenient observation and operation of all items by an operator approximately 5 feet 10 inches tall.

3.14.1.7.2 Grouping of controls and gages

3.14.1.7.2.1 No. 1 system. All controls and gages for the No. 1 system shall be grouped in one section of the control panel. Valve packing nuts, handles, buttons, and gage face rings shall be painted red.

3.14.1.7.2.2 No. 2 system. All controls and gages for the No. 2 system shall be grouped in one section of the control panel. Valve packing nuts, handles, buttons, and gage face rings shall be painted green.

3.14.1.7.2.3 No. 3 system. All controls and gages for the No. 3 system shall be grouped in one section of the control panel. Valve packing nuts, handles, buttons, and gage face rings shall be painted blue.

3.14.1.7.2.4 Color coding of panel. In lieu of color-coding each nut, handle, button, and gage face ring to distinguish systems, the control panel may be color coded.

3.14.1.7.2.5 The fill button, fill system bleed valve pushbutton, and sight tube shall be grouped together. The air compressor controls and gages shall be grouped together on one section of the control panel.

3.14.1.8 Wheels and tires. The wheels and tires shall be selected in accordance with MIL-M-8090.

3.14.2 Housing. The test stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 U.S. gage (0.0598 inch) or heavier, sheet steel with doors and adjustable louvers as required. The housing shall be designed to support a 200-pound weight distributed evenly over an area 4.5 by 5 inches and applied in any location on the top of the test stand. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the stand or handling of the housing while removed from the stand. The test stand shall be so designed that water or hydraulic fluid spilled any place within the housing will drain to the exterior through one outlet and will not collect within the enclosure. Metal hooks shall be provided on the exterior of the housing to permit temporary storage of the pressure hoses. These hooks shall be designed to permit flat positioning against the housing when not in use.

3.14.2.1 Internal accessibility and ventilation. The housing shall be provided with access and ventilating doors and adjustable louvers to permit access for operating and servicing and to

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ventilate the stand. The doors and louvers shall be free from binding and shall be easy to operate with one hand. Provisions shall be incorporated to rigidly secure the doors in the fully opened or fully closed position.

3.14.2.2 The housing shall be easily removable by actuating not more than 10 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the stand. Handles shall be provided for this purpose.

3.14.2.3 Reservoir. A stainless-steel or aluminum hydraulic-fluid reservoir shall be mounted within the housing. The reservoir shall have adequate capacity to supply at least 25 gallons of fluid to the system when inclined at 8-1/2° from true vertical in any direction. The reservoir shall have additional space for a 3-gallon expansion above the normal full level and at least a 2-gallon space below the outlet to the system for collecting moisture and impurities.

3.14.2.3.1 The reservoir shall conform to MIL-R-5520 regarding obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, return line inlet, scuppers, filler unit, finish, and filling time. Clearance shall be provided around the filler opening to allow rapid filling from 1-gallon containers while the stand is in operation.

*3.14.2.3.2 Access plates. One or more access plates at least 7 inches in diameter shall be provided to permit visual inspection and manual cleaning of the interior of the reservoir.

3.14.2.3.3 Drain plug. A drain plug shall be provided for completely draining the reservoir. Clearance shall be provided around the drain outlet to permit rapid draining into 1-gallon containers.

3.14.2.3.4 Fluid level indicator. A dial-type fluid level indicator shall be provided and shall indicate the available fluid in the reservoir to an accuracy within ±2 gallons of actual quantity. The indicator shall be mounted on the instrument panel.

*3.15 Compartments. A handbook compartment, 2 by 9 by 12 inches or larger, and an enclosed accessory box, 8 by 8 by 24 inches or larger, shall be permanently attached at an accessible point within the housing. The compartment shall be constructed on the trailer bed or shall be fastened to the housing assembly. Provisions shall be made for storage of the hose assemblies and accessories specified herein within the outline of the housing.

3.16 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.17 Electromagnetic interference. The test stand shall meet the requirements of MIL-STD-826, method 7001.

3.18 Noise level. The noise level of the test stand shall not exceed the values in the frequency ranges specified in 4.6.17.

3.19 Dimensions. The length and width of the test stand and housing shall be not less than the trailer frame assembly nor extend beyond the trailer wheels at any point. The overall height of the test stand shall be held at a minimum consistent with good design practices, but shall

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not exceed 44 inches above the top of the tires.

*3.20 Weight. The weight of the test stand, including the trailer assembly, shall not exceed 6,000 pounds.

3.21 Finishes and protective coatings. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. Tubing and components that have been treated with corrosion-resistant coatings need not be painted. All exterior surfaces and the inner surfaces of the doors shall be painted yellow, film designation DG. All other interior surfaces shall be painted gray, film designation LG.

3.22 Operational markings

3.22.1 An engraved or etched, plastic or metal nameplate containing the following information shall be permanently attached to each pressure outlet:

PRESSURE OUTLET NO. _____.

3.22.2 An engraved or etched, plastic or metal nameplate containing the following information shall be permanently attached to each suction connection outlet:

SUCTION RETURN NO. _____.

3.22.3 An engraved or etched permanent-type temperature correction chart shall be installed near the flowmeters.

3.22.4 The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be in 1/4-inch black letters and shall be visible to the operator.

3.22.5 The following information shall be marked in 1-inch black letters on the tow bar and the front of the housing:

MAX SPEED 20 MPH.

*3.22.6 Instrument panel markings

3.22.6.1 All engine and heater instruments and controls shall be identified by etched or engraved, metal or plastic plates permanently attached to the instrument panel.

3.22.6.2 An 8- by 10-inch engraved, permanent-type diagram plate, arranged similar to figure 1 to show the hydraulic circuit of the stand, shall be securely fastened to the instrument panel.

3.22.7 Control panel markings

3.22.7.1 A permanently engraved plate containing low-temperature-adjustment directions for the oil cooler(s) shall be provided on the control panel.

3.22.7.2 All items mounted on the control panel shall be identified as to use and cautions by etched or engraved, metal or plastic plates permanently attached to the control panel.

3.22.8 Stenciling

3.22.8.1 The fuel tank cap shall be stenciled to denote the type of fuel to be used.

3.22.8.2 The following information shall be stenciled in 1-inch black letters on each of the four sides of the test stand:

FOR AIR SHIPMENT, THIS STAND IS DESIGNED FOR 8G FORE AND AFT, 3G SIDEWARD.

3.22.8.3 The exterior of the test stand shall be stenciled to indicate the pressure,

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center-of-gravity, and other markings necessary for proper handling and maintenance.

3.23 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional marking shall be included:

Rating: Type TTU-228/E

Three systems operating:

0-20 gpm at 3,000 psi each system
0-10 gpm at 5,000 psi each system

Two systems operating:

0-30 gpm at 3,000 psi each system
0-15 gpm at 5,000 psi each system.

*3.24 Government-furnished property. When the contract or purchase order so provides (see 6.2), the Government will furnish one MS24560-1 gasoline engine to the contractor upon his request, for use in fabricating the test stand.

3.25 Workmanship. The test stand, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts; et cetera.

3.25.1 Riveting. Riveting operations shall be performed to insure tight and satisfactory heading of rivets.

3.25.2 Cleaning. The test stand shall be thoroughly cleaned. Loose, spattered, or excess solder; metal chips; and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test stand shall be classified as follows:

a. Preproduction testing. . . . . See 4.4

b. Acceptance tests . . . . . .See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Test time. Unless otherwise specified, any test period shorter than 1 hour shall not be credited as test time.

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4.3.3 Apparatus and instruments. Apparatus and instruments used in conjunction with the testing specified herein shall be of laboratory precision-type and shall be checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation. They shall be calibrated at intervals properly spaced to insure laboratory efficiency.

4.3.4 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with a certified accuracy of 1 percent or better of reading.

4.3.5 Preliminary run-in. Any preliminary run-in of the test stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the test stand for testing. A certified log of all preliminary running time, calibration data, et cetera, shall be included in the test report.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One test stand shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plan and tests.

4.5.1 Individual tests. Each test stand shall be subjected to the following tests as described under 4.6:

a.	Examination of product	See 4.6.1
b.	Calibration test	See 4.6.2
c.	Flow tests	See 4.6.3
d.	Loading pressures	See 4.6.4
e.	Fluid temperature test	See 4.6.5
f.	Road test	See 4.6.25.

4.5.2 Sampling plan and tests

4.5.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.5.2.1.1 MIL-STD-105. When MIL-STD-105 specifies an action by the Government it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.5.2.2 Tests. Samples selected shall be subjected to the following tests.

a.	Semi-endurance test	See 4.6.6
b.	Maneuverability tests	See 4.6.26.

4.6 Test methods

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with requirements specified herein with respect to parts, accessibility, ease of maintenance, materials, workmanship, and markings.

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4.6.2 Calibration test. The pressure drop from the outlet of the high-pressure pump to the end of the 30-foot pressure hose and the pressure drop from the end of the 30-foot return hose to the suction side of the boost pump shall be measured and shall not exceed 40 psi on the pressure side or 20 psi on the return side.

*4.6.3 Flow tests. The constant-flow operation of each system of the test stand shall be checked at 800, 1,500, and 3,000 psi at each rate of flow of 6, 12, 20, and 30 gpm from each outlet. The pressure gages shall be accurate within ±1 percent of full scale range at all settings. The test flowmeters shall be accurate within ±1 percent of full scale reading. Each system of the test stand shall then be operated at 10 gpm and 5,000 psi for 1 hour and inspected for evidence of leakage of fluid.

4.6.4 Loading pressures. The operation of each system of the test stand shall be checked at 1,500 and 3,000 psi at each rate of flow of 15 and 30 gpm from each outlet and with compensation set at 1,500 and 3,000 psi, respectively. The hydraulic pumps shall put out full 15 and 30 gpm flow at all loading pressures below the 1,500 and 3,000 psi respectively, but flow shall fall to nearly 0 gpm as soon as the loading pressure reaches the compensation pressure setting.

4.6.5 Fluid temperature test. The temperature of the fluid entering each high-pressure pump on the test stand shall be raised above 160°F and the thermoswitches shall sound the warning horn before the fluid temperature exceeds 165°F but not before it reaches 160°F. Each system shall be checked independently. The warning horn shall be audible in a radius of 25 feet from the test stand in any direction.

4.6.6 Semi-endurance test. The test stand shall be subjected to 20 percent of the endurance test specified in 4.6.9. Upon completion of this test, the test stand shall satisfactorily pass the tests specified in 4.6.3 and 4.6.4.

4.6.7 Initial disassembly inspection. The test stand shall be disassembled to the maximum practical extent and completely inspected, the breakdown weights determined, and measurements made which may be used to determine wear. All gages, meters, and instruments shall be laboratory checked against certified, calibrating equipment and failure to meet the accuracy requirements specified for any item shall be cause for rejection. All requirements which may be determined by weight, torque and force requirements, dimensional measurements, and inspection shall be checked for compliance, and failure to meet any requirement herein shall be cause for rejection. Following this inspection, the test stand shall be reassembled and subjected to the remainder of the preproduction tests.

4.6.8 Environmental tests. The test stand shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810. Unless otherwise specified, after completion of each test, the test stand shall be subjected to the tests specified in 4.6.3 and 4.6.4. Failure of the test stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition which cannot be remedied by routine maintenance shall be cause for rejection.

4.6.8.1 Low temperature. The test stand shall be subjected to low temperature in accordance with method 502,

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procedure I, except the storage temperature shall be -65°F. The test circuit shall include three high pressure hoses, restrictor valves to build up 3,000 psi pressure, the reservoir filled with not less than 25 gallons of hydraulic fluid conforming to MIL-H-5606, and three MS28741-24-1200 return lines. The entire test circuit shall be contained within the cold chamber and the stand shall not be started until after the temperature has stabilized at -40°F. The fluid in the reservoir shall not be heated prior to starting or during the test from any external source. The engine shall start after 30 minutes or less of preheating. After starting, the flow and pressure of each system shall be slowly increased to the rated output of 20 gpm and 3,000 psi. The test stand shall attain rated output within 30 minutes after starting and shall be operated at rated output for not less than 30 minutes. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation in the -40°F ambient temperature. The high-pressure gage shall be accurate within ±1/2 percent of full scale range.

4.6.8.2 High temperature. The test stand shall be subjected to high temperature in accordance with method 501, procedure I. Upon completion of the exposure to high temperature, the ambient temperature shall be lowered to 125°F. The constant flow operation of the stand shall be checked at 800, 1,500, and 3,000 psi at each flow rate of 6, 12, and 20 gpm for each system. The high-pressure gage shall be accurate within ±1/2 percent of full scale range at all settings. The flowmeters shall be accurate within ±1/2 gpm of actual flow. The temperature shall then be reduced to 100°F. Each of the three systems shall be operated at 20 gpm at 750 psi through the bypass circuit for 1 hour. The test stand cooler shall maintain the fluid temperature below 140°F. After this test, the test stand shall be inspected for leakage of fluid.

4.6.8.3 Fungus resistance. The test stand shall be subjected to fungus in accordance with method 508, procedure I.

4.6.8.4 Sand and dust. The test stand shall be subjected to sand and dust in accordance with method 510, procedure I.

4.6.8.5 Humidity. The test stand shall be subjected to humidity in accordance with method 507, procedure I.

4.6.8.6 Salt-fog. The test stand shall be subjected to salt-fog in accordance with method 509, procedure I.

4.6.8.7 Rain. The test stand shall be subjected to rain in accordance with method 506, procedure I. The test stand shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 1 hour and while still in the rain test, the test stand shall be started and three systems operated at 20 gpm and 3,000 psi for 30 minutes. Failure of the test stand to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces which might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 inch deep or which could be detrimental to the performance of the stand shall be cause for rejection.

4.6.9 Endurance test. The three systems of the test stand shall be connected in series to a test rig through 30 feet of pressure hoses and 30 feet of suction hoses specified herein. The test rig shall include a reservoir, restrictors to create test pressures in the high-pressure lines,

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restrictors to create the prescribed vacuum in the return lines, and a cooler(s) to maintain fluid temperatures at approximately 130°F. The ambient air temperature shall be maintained at 75°F or higher during the endurance test. Readings shall be recorded every hour of all gages, meters, and instruments on the stand. Output flow, fluid temperature, fluid pressure, inlet fluid temperature and pressure, amperes, and total gasoline consumed shall be recorded each hour (see 4.6.10). Endurance test runs shorter than 1 hour shall not be credited as part of the endurance test. The test stand shall first complete the test specified in 4.6.3 and shall then be subjected to the following 350 hours of endurance testing. The fill motor and pump shall be operated continuously during this period.

a. One hundred hours of operation of each system delivering a combined flow of 54 gpm and 3,000 psi. Pressure shall be discharged through the 1-inch outlets and returned through the 1-1/2-inch inlets with suction pressures of 125 psig.

b. Fifty hours of operation at 10 gpm from each system and 5,000 psi. Pressure shall be discharged through the 1-inch outlets and returned with suction pressures of 0 to 10 psig.

*c. One hundred hours of operation of each system delivering a combined flow of 54 gpm and 3,000 psi with suction pressure to the low-pressure pump reduced to 20 inches Hg absolute. Pressure shall be discharged through the one 1-inch outlet and returned through another 1-1/2-inch inlet.

*d. Fifty hours of operation at 10 gpm from each system and 1,000 psi with suction pressures of 20 to 30 inches Hg absolute.

e. Fifty hours of operation at 10 gpm from each system and 1,000 psi with suction pressures of 125 psig.

4.6.9.1 During the endurance tests, the flow of each system shall be instantaneously varied from the preset flow to zero flow and immediately returned again to the preset flow. This cycle shall be accomplished one thousand times during each 50 hours of testing. Any damage to the pumps, pump compensators, flow indicators, gearing, or other components shall be cause for rejection.

4.6.9.2 The test specified in 4.6.3 shall be conducted. This data shall be compared with data recorded in 4.6.9. There shall be no indication of failure or undue wear to the test stand.

4.6.10 Air compressor test. The air compressor shall be operated at least 25 percent of the time during the endurance test specified in 4.6.9. The performance of the compressor shall be checked periodically during each test. Failure of the drive coupling or any marked decrease in performance or failure of a component part of the pressurizing system shall be cause for rejection. The output of the compressor shall be determined and shall equal or exceed the requirements specified in 3.11.

4.6.11 Pump endurance tests. The high-pressure pumps shall be driven at rated rpm for a minimum of 1,000 hours as specified under 4.6.9. Unless otherwise specified, fluid temperatures during the 1,000 hours shall be maintained at 80°F 25 percent of the time for each flow condition.

4.6.12 Compensation time. Inlet pressure, outlet pressure, and delivery of the pumps shall be recorded at one-half

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flow and maximum pressure, rated flow and pressure, and rated flow at one-third pressure to show compensating characteristics of the pumps. The pumps shall compensate from zero flow to full flow and vice versa in 0.05 second, and the pressure shall not exceed 133 percent of the selected pressure.

*4.6.13 Low-pressure pump tests. When low-pressure pumps are used, they shall be tested at rated flow and speed for 1,000 hours under the following conditions:

Running Time (Hours)	Inlet Conditions (Inches Hg Abs)	Outlet Pressures (Min)
300	20	Maximum rated pressure
200	25	Three-fifths maximum rated pressure
300	15	Two-fifths maximum rated pressure
100	20 to 30	0.44 maximum rated pressure
100	20	Maximum rated pressure.

4.6.14 Fill pump tests. The fill pump shall be operated with the gasoline engine on the test stand stopped. The fill pump shall deliver 2 gpm against a pressure of 150 psi.

4.6.15 Altitude. A test report shall be prepared with calculations showing that the engine used with the stand will drive all the test stand components at full rated capacity while operating at a pressure altitude of 6,000 feet, 130°F. The following SAE correction formula shall be used in this calculation:

$$C.F. = \frac{29.92}{23.98} \times \frac{590}{520.}$$

4.6.16 Electromagnetic interference tests. The test stand shall be tested as a complete unit for radiated interference in accordance with MIL-STD-826, method 7001.

4.6.17 Noise level test. The noise level of the test stand shall be measured 10 feet from the perimeter of the test specimen while each system of the test stand is operating at 20 gpm and 3,000 psi. The test stand shall not exceed the noise level shown below. Measurements shall be taken out-of-doors on a paved surface from at least eight different directions, 45° apart, in a horzontal plane parallel to the trailer bed and passing through the centerline of the high-pressure pump drive-shaft.

Cycles Per Second	Not Exceeding (db)
20 to 75	108
75 to 150	98
150 to 300	90
300 to 600	83
600 to 1,200	78
1,200 to 2,400	75
2,400 to 4,800	72
4,800 to 10,000	71.

4.6.18 Valve tests. The flow control valves shall be tested by connecting the test stand outlets to external flowmeters. With the pumps set to operate at 3,000 psi and 30 gpm, each flow control valve shall be operated to determine that they can hold the outlet flow steady at any

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desired value from 0 to 30 gpm. Failure of the valves to regulate the flow to 1/4 gpm at any setting from 0 to 5 gpm and to 1 gpm at any setting from 5 to 30 gpm shall be cause for rejection. The flow control valves shall be tested for leakage with the pumps operating at 3,000 psi. The force required to turn the valves under the above conditions shall not exceed 40 pound-inches. Any leakage shall be cause for rejection. During these tests, the flow control valve settings shall not be influenced by vibration from the test stand.

4.6.19 Stand operation (tilted position). The complete test stand shall rest on a platform and shall be kept from rolling by the parking brakes. The platform supporting the stand shall be tilted in eight different directions, 45° apart, on an 8-1/2° incline with the true horizontal plane. The test stand shall be started and operated at 54 gpm and 3,000 psi for 30 minutes in each position. The test stand shall perform satisfactorily in every respect, and the parking brakes shall prevent the stand from moving on the platform. While tilted in each of the eight positions, 25 gallons of fluid shall be pumped from the reservoir and then replaced to check the capacity specified in 3.14.2.3.

4.6.20 Support test. A 200-pound weight shall be applied slowly and uniformly to an area 4-1/2 by 5 inches in at least six random positions on top of the test stand housing, including a reading in the center of each panel. The test stand shall not be permanently deformed by the application of weight.

4.6.21 Mobility tests. The test stand shall be subjected to and shall pass all tests specified in MIL-M-8090 for type II, group C mobility. Upon completion of the mobility tests, the tests specified in 4.6.3 and 4.6.4 shall be repeated. Calibration of gages and minor adjustments will be permitted during the tests.

4.6.22 Serviceability. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, which may hinder servicing and operation.

4.6.23 Final disassembly inspection. The test stand shall be disassembled to the maximum practical extent and a thorough examination of all parts conducted to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. If measurements were made to determine wear, they shall be compared with similar measurements made during initial dissassembly inspection. Test stands not conforming to requirements specified herein shall be rejected.

4.6.24 Differential pressure indicator. No. 1, No. 2, and No. 3 high-pressure filter differential indicators shall be tested for proper operation. The differential indicators shall come on when the differential pressure across the filters is 100 ±10 psi. The indicators shall operate at any line pressure between 750 and 5,000 psi.

4.6.25 Road test. The test stand shall be towed for a distance of 10 miles to determine general functioning of running gear components. The trailer shall be inspected for excessive vibrations, distortion, loosening, and maladjustment during and following this test. Evidence of malfunctioning shall be cause for rejection.

4.6.26 Maneuverability tests. The test stand shall be subjected to the following tests. During and following these tests, the test stand shall exhibit no evidence of failure or impending failure.

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4.6.26.1 The test stand shall be towed 100 miles over improved roads (see 6.4.3) at an average speed of 20 mph, maximum speed of 25 mph; and not less than 50 miles over graded gravel road at an average speed of 10 mph, maximum speed of 12-1/2 mph.

4.6.26.1.1 Not less than 20 sudden stops shall be made while towing at each speed specified in 4.6.26.1.

4.6.26.2 The test stand shall make not less than 20 full turns to the left and 20 full turns to the right at 8 mph at its minimum turn radius.

4.6.27 Air transportability. The test stand shall be subjected to the air transportability test specified in MIL-A-8421. The test stand shall be secured during this test by means of the lifting and tiedown fittings only. Upon completion of this test, the test stand shall be inspected to determine adequacy of bracing, adequacy of tiedown and liftings, installation, and security of mounting of all components. The test stand will not be required to be operable upon completion of the 8-g acceleration loading. If it can be shown by computation that the test stand and all components will withstand the 8-g acceleration loading, the computation may be substituted in lieu of the test.

4.6.29 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.30 Maintainability evaluation. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The test stand shall be preserved and packaged in accordance with method I of MIL-P-116.

5.1.1 Cleaning and drying. All surfaces to which a contact preservative is to be applied shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Preservative application. All bare metal surfaces subject to rust or corrosion shall be coated with preservative conforming to type P-2 of MIL-P-116. Unpainted information plates shall be coated with ignition insulation compound conforming to MIL-V-1137.

5.1.1.3 Engine and compressor. The engine and compressor shall be preserved in accordance with MIL-E-10062.

5.1.1.4 Hydraulic system. The hydraulic fluid shall be drained from the hydraulic system and refilled to its full operating capacity with preservative oil conforming to MIL-H-6083. A tag conforming to

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UU-T-81 containing the following information shall be attached to the system:

CAUTION

Drain the preservative oil conforming to MIL-H-6083 from the hydraulic system and refill with hydraulic fluid conforming to MIL-H-5606 before placing the equipment in operation.

5.1.1.5 Electric motor. The electric motor shall be preserved and packaged in accordance with MIL-P-16298.

5.1.1.6 Vent openings. All vent openings shall be covered with barrier material conforming to MIL-B-121, grade A, and secured in position with tape conforming to PPP-T-60.

5.1.1.7 Hose and electrical cable. Hose ends shall be covered with barrier material conforming to MIL-B-121, grade A, and secured in position with tape conforming to PPP-T-60. The hose and electrical cable shall be coiled to a safe minimum diameter and securely bound in three places with cotton tape. The bound coils shall be secured on the inside of the test stand.

5.1.1.8 Drawbar. The drawbar shall be removed and secured to the stand. It shall be readily removable to facilitate handling and shipping operations.

5.1.1.9 Shrouding. The test stand shall be covered with a shroud fabricated of material conforming to MIL-C-20696. The shroud shall be provided with adequate tiedown facilities.

5.1.2 Level C. The test stand shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A, B, or C. The test stand preserved and packaged as specified in 5.1.1 or 5.1.2 shall be shipped mobile (uncrated).

5.3 Marking. In addition to any special marking required by the contract or order, the test stand shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Test Stand, Aircraft Hydraulic System TTU-228/E, Trailer-Mounted, Gasoline-Engine-Driven, Triple System.

6. NOTES

6.1 Intended use. The TTU-228/E test stand is intended for use in ground checking and maintenance of aircraft hydraulic systems. The test stand provides a total of three systems of 54 gpm flow at 3,000 psi or 30 gpm flow at 5,000 psi.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether a MS24560-1 gasoline engine is to be furnished to the contractor (see 3.24)

c. Location and condition for preproduction testing (see 4.4)

d. Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).

6.2.1 Gasoline engine tests. If the gasoline engine is furnished by the contractor, it should be subjected to the

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qualification tests specified in MIL-E-26248 or it should be shown to be a qualified product in accordance with MIL-E-26248.

6.3 The high-pressure pumps on the TTU-228/E test stand are mounted on standard accessory drive pads. Aircraft pumps can be substituted by the use of the following parts:

AN4139	Adapter-Type XVI Drive Flange
AN4182	Adapter-Type XVII Drive Spline.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{MTBF}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.4.2 Mean-time-betwee-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.4.3 Improved road. An improved road is a smooth, hard-surfaced road such as a concrete or asphalt-paved highway.

6.4.4 Dissassembled to the maximum practical extent. Disassembly to the maximum practical extent is defined as disassembly of all moving contacting surfaces which have been assembled by the manufacturer of the stand and which are subject to wear (gears, bearings, control mechanisms, et cetera).

* 6.5 International agreements. Certain provisions in 3.6, 3.9.1, 3.9.1.4, and 3.9.3 of this specification are the subject of international standardization agreement (American-British-Canadian Air Standardization Coordinating Committee Air Standard 11/14). Certain provisions in 1.1, 3.6, 3.9.1.4, 3.9.3, 3.9.18, and 3.22.6.2 of this specification are the subject of international standardization agreement (North Atlantic Treaty Organization Standardization Agreement 3510).

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity:
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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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